

Zijun Liu

17560840848 | zijunliu21@gmail.com



EDUCATION

University of Chinese Academy of Sciences	2021.08 - 2024.06
<i>Master of Biological and pharmaceutical engineering</i>	
Shandong Management University	2017.08 - 2021.06
<i>Bachelor of Economics</i>	

Research Experience

1. Investigating Neural Circuit Mechanisms of Social Status-Modulated Addiction Vulnerability	
<i>Independent Researcher</i>	Jul 2022 – Jun 2024
- Engineered chronic nicotine addiction models in mice and stratified susceptibility based on withdrawal-induced craving; established social hierarchies using reward competition and tube test paradigms. - Automated behavioral annotation by integrating DeepLabCut (DLC) for multi-agent pose tracking with SimBA-based supervised machine learning, enabling precise quantification of hierarchy-related behaviors (e.g., chasing, lateral threats). - Identified a robust negative correlation between social rank and nicotine addiction vulnerability across cohorts. - Decoded neural circuits using optogenetics and multi-channel fiber photometry, pinpointing the VTA→mPFC dopamine pathway (rather than VTA→NAc) as the core driver of competitive motivation; reversed competitive deficits in subordinate mice via pathway-specific phasic activation. - Elucidated the molecular substrate by demonstrating that D2 receptor antagonists block hierarchy-elevating effects; validated significantly higher D2R expression in the mPFC of dominant mice via Western blot analysis.	
2. Systematic Evaluation of Nicotine Withdrawal via a Hierarchical 3D Motion Learning Framework	
<i>Lead Investigator</i>	Feb 2023 – Apr 2025
- Spearheaded high-dimensional behavioral phenotyping using the BehaviorAtlas 3D motion learning framework, systematically extracting sub-second behavioral motifs associated with withdrawal phases and individual susceptibility. - Developed a digital evaluation pipeline utilizing Support Vector Machines (SVM) and regression models, achieving >94% accuracy in automated withdrawal diagnosis and high-precision quantification of withdrawal severity ($R^2 > 0.65$), establishing a robust alternative to manual scoring. - Discovered and defined a novel "stereotypic wandering" withdrawal cluster via UMAP manifold mapping, correcting the traditional bias of withdrawal as mere "immobility" and revealing concurrent dynamics of somatic pain and psychomotor agitation. - Pioneered a high-throughput pharmacological screening platform to objectively quantify the efficacy of Varenicline and novel therapeutics in suppressing pathological behaviors and restoring basal exploration.	
3. Impact of Nicotine Administration Routes on Addiction Modeling and Behavioral Phenotypes	
<i>Lead Investigator</i>	Jul 2022 – May 2024
Evaluated the heterogeneous impacts of three administration routes (smoke exposure, IP injection, oral water) on cognitive function, anxiety levels, social hierarchy, and aging progression in mice.	

- Tracked dynamic physiological parameters (weight, metabolism, motor function) in real-time, providing a standardized reference framework for addiction model selection.

Independent Proposal: Neural Dynamics of Social Hierarchy – An Energy Landscape and Attractor Framework

Independent Research & Design

Oct 2025 – Nov 2025

- Formulated the "Heterogeneous Energy Landscape" hypothesis, conceptualizing social hierarchy formation as a physical evolution of mPFC neural dynamics from a "fluid state" to a "solidified state."
- Implemented recurrent Switching Linear Dynamical Systems (rSLDS) to decode latent strategy states and defined an "Attractor Stability Score (ASS)"; proposed integrating optogenetic perturbations with intrinsic timescale analysis to quantitatively validate how social experience reshapes the depth of mPFC attractor basins.

TECHNICAL SKILLS

1. Computational Neuroscience & Data Mining

- Programming & Environment: Proficient in Python (NumPy, Pandas, SciPy, Scikit-learn) and bioinformatics (Scanpy, pySCENIC); adept at deploying analytical environments and building efficient data pipelines in Linux.
- Calcium Imaging Processing: Mastered Suite2p and CaImAn (CNMF-E) for end-to-end processing of 2-Photon/Miniscope data, including motion correction, ROI extraction, and signal denoising.
- Electrophysiology Analysis: Skilled in pyabf for sampling data preprocessing and SpikeInterface for high-throughput Neuropixels spike sorting and downstream analysis.
- Neural Dynamics Modeling: Utilized the ssm library to construct generative models (rSLDS); adept at PCA, UMAP, and Neural Velocity Field analysis for quantitative manifold mapping of high-dimensional population trajectories.
- Advanced Algorithms: Exploring Graph Neural Networks (GNN) for multi-agent topological modeling and Transformer architectures for decoding long-term dependencies in neural populations.

2. Precision Behavioral Tracking & Quantification

- Pose Estimation: Engineered automated tracking pipelines using DeepLabCut / SLEAP combined with SimBA, enabling robust quantification of complex social interactions.
- 3D High-Dimensional Parsing: Deployed BehaviorAtlas / MoSeq frameworks for 3D, multi-keypoint, sub-second pose estimation, extracting objective kinematic features of innate behaviors.

3. In Vivo Manipulation & Wet-Lab Expertise

- Circuit Manipulation: Proficient in stereotaxic surgeries, viral microinjections (e.g., transsynaptic tracing), and in vivo optogenetics / chemogenetics (DREADDs).
- Neural Recording: Extensive hands-on experience with multi-channel fiber photometry data acquisition.
- Histology: Mastered standard procedures including transcatheter perfusion, brain slicing, Immunohistochemistry (IHC), and confocal microscopy.

PUBLICATIONS

Zijun Liu, Mengke Su, Shuhui Jia, Zhibin Xu, Kang Huang, Zhizhun Mo, Yaning Han, Xinan Liu, et al. "Systematic evaluation of nicotine addiction in mice using a hierarchical 3D-motion learning framework." (*First Author, Under Review*)